



ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTTP/ELECT MAINT/2024-25/SWYDESP/164929

Date:25/05/2024(18:14:34)

Subject:Revision of LMI Monitoring the healthiness of 400KV CT

Reason for Revision of LMI/OIN/OPS/ELEC/035: OIN/ELEC/035 on which our LMI is made is revised in Mar 2024.

Changes done in present revision: Mentioned in LMI clause 4.0 point b (pg1).

Attachments:

- 1) Revised LMI
- 2) Old LMI
- 3) OIN-ELEC-35
- 4) OIN-ELEC-42.

Competent authority for approving the LMI revision is HOP Simhadri. Kind approval may be accorded.

Submitted by:

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Note No:5

please modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

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Note No:6

Modification done as per standard format.

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Note No:7

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Note No:8

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Note No:9

please review and modify the LMI format as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04

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Note No:10

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On:24/06/2024(10:30:04) **Ref:**SMTTPP/ELECT MAINT/2024-25/SWYDESP/164929

Note No:11

Done as per standard Simhadri format

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Note No:12

may be processed

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Note No:13

Bring out the revisions in the proposal sheet

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Note No:14

Changes in previous revisions is in old LMI.

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Note No:15

May please approve

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Note No:16

Please discuss

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Note No:17

Changes done in present revision

- i) EHV equipment failure format revised in-line with guideline mentioned in CEA document. Format mentioned in 8.0 i.e Record keeping.
- ii) Special instructions to be followed removed as it is repeat of maintenance schedule and acceptance norms.
- iii) Maintenance schedule along with formats and acceptance norms and its annexures added as per OIN-OPS-ELEC-035 and OIN-OPS-ELEC-042

Submitted by:

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Note No:18

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Note No:19

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On:16/07/2024(10:13:34) **Ref:**SMTPP/ELECT MAINT/2024-25/SWYDESP/164929

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Monitoring healthiness of 400KV Current Transformer

LMI/OIN/OPS/ELEC/035 Rev: 05

MAY- 2024

Approved for implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	MONITORING THE HEALTHINESS OF 400KV CURRENT TRANSFORMER
2	LMI NO	LMI/OIN/OPS/ELEC/035 Rev: 05
3	Reference document	O&M Manual of 400KV CT, OIN-OPS-ELEC-035, OIN-OPS-ELEC-042
4	Superseded document	LMI /OIN/OPS/ELEC/035 Issue: 1 Rev: 4 dated: July 2023.
5	Prepared by	Name: Anil Kumar Pinninti, SR. MANAGER(EM)
6	Checked by	Name: D.K. Tomar, AGM I/C (EM)
7	Reviewed by	Name: Suresh John David, GM (O&M)
8	Approved by	Name: S.K Sinha, HOP (Simhadri)

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TITLE	MONITORING THE HEALTHINESS OF 400KV CURRENT TRANSFORMER
LMI No.	LMI/OIN/OPS/ELEC/035 REV:5

1.0 OBJECTIVE:

400 KV Current Transformers are used to measure the current for metering and protection purpose. There are total 63 no's of 400 KV CTs supplied by GE T&D in STAGE#1 and 78 Nos of 400 CTs supplied by CGL in STAGE#2. The healthiness of the CTs is to be ensured for safe and reliable power transmission to the grid. EHV grade transformer oil is used in the 400KV CTs for insulation and cooling. CT healthiness is checked by the following PDM techniques.

1. DGA
2. Tan Delta and Capacitance Measurement.
3. IR measurement.
4. Bellow position monitoring.

SCOPE:

The document covers monitoring the healthiness of all the 400 KV CTs by

1. Monitoring of DGA trends of oil.
2. Measurement of Tan delta and Capacitance value.
3. Insulation resistance measurement.
4. Regular bellow position monitoring (for CGL & GE make live tank CTs)
5. Periodical Checks to be done on daily, monthly, and annual / biennial maintenance.

2.0 SUPERSEDED DOCUMENTS :

- a. LMI /OIN/OPS/ELEC/035 Issue: 1 Rev: 4 dated: July 2023.

3.0 RECORD OF REVISION:

- i) EHV equipment failure format revised in-line with guideline mentioned in CEA document. Format mentioned in 8.0 i.e Record keeping.
- ii) Special instructions to be followed removed as it is repeat of maintenance schedule and acceptance norms.
- iii) Maintenance schedule along with formats and acceptance norms and its annexures added as per OIN-OPS-ELEC-035 and OIN-OPS-ELEC-042

4.0 MAINTENANCE SCHEDULE ALONG WITH FORMATS & ACCEPTANCE NORMS

The maintenance schedules and acceptance norms are placed at Annexures 3,4 & 5. Formats of CT monitoring & maintenance records for daily, monthly, yearly and once in two Year maintenance are provided in Annexures-6,7,8.

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5.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Switchyard area maintenance engineer for implementation and overall responsibility lies with sectional head of electrical maintenance.

6.0 EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS

Any exceptions / deviations shall be reported to HOD (EM). And failure of CT to be reported as per annexure-9 format.

7.0 RECORD KEEPING:

- Records will be maintained by Switchyard area maintenance engineer as per annexure-6,7,8 as per maintenance schedule annexure-4.
- Any EHV equipment failure report shall be submitted to COS-Elec for further submission to CEA as per below mentioned format (Annexure-9)**

8.0 REVIEW:

This LMI document will be reviewed once in every two years by Head of the Project, NTPC Simhadri.

9.0 Annexures:

Annexure-1 Common Failure mechanisms associated with CT

Sr. no	Component	Failure cause	Main effect	Secondary effect	Failure mode
1	Terminal connection	looseness	Heating /hotspot at joint	Damage to oil gasket. Leakage of oil from bellow/ gasket joint. Breathing of air and moisture into the oil	Increased dielectric heating causing thermal runaway.
2	Oil seals	Ageing of oil seal Improper material	- Oil leakage N2 leakage	Breathing of air and moisture into the oil Breathing of air and moisture into the oil	Increased dielectric heating causing thermal runaway.

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3	Primary Insulation	- Defect in paper/foil wrapping. - Presence of sharp edges, voids/wrinkles in paper / foil layers - Poor drying and oil impregnation - Foreign material in insulation (dirt, debris etc)	- Incipient ionization leading to partial discharge. - Thermal runaway - Accelerated ageing/degradation	- puncturing of insulation - Earth fault - Dielectric failure	Partial discharge leading to dielectric failure
4	Magnetic core	- Shorting of core insulation - Overheating of core	Hot spot generation in oil	- Damage to primary paper insulation and oil - Generation of gas in oil	- Dielectric Failure -High DGA
5	Secondary winding	- Open circuiting of CT secondary	- Overheating of core - Shorting of Secondary winding	- Damage to primary paper insulation and oil - Generation of gas in oil	- Dielectric failure - High DGA
6	PF test cap/connection	- Pitting at earthing point -Loss of grounding	- Sparking at PF terminal - Improper voltage grading in OIP Capacitors.	Failure of Primary insulation	- Dielectric failure
7	Oil	- Contamination with moisture and over saturation with O₂,N₂ - Ageing of oil - Rapid lowering of temperature - Presence of aromatic content in oil	- Increase in tan delta - Increase in tan delta - Evolution of bubbles and PD appearance. - Accelerated ageing	- increased heating of oil and winding - increased heating of oil and winding - PD activity due to over saturation of oil - increased heating	- Thermal run away -Dielectric failure - Dielectric failure - Thermal run away

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Annexure-2 Current Transformer Failure Modes

Sr. No	Cause	Effect	Failure mode	Remedial measures
1.	Poor Primary bar Clamping arrangement design	May fail during short circuit condition. Mechanical shifting of hairpin could lead to damage to primary insulation and its failure.	Mechanical	Adequacy of clamping arrangement to withstand short circuit forces to be ascertained during design review.
2	Poor PF terminal connection design	Hotspot generation on PF terminal, causing pitting and floating voltage across condenser layers may cause primary insulation failure.	Dielectric failure	Proper design of test tap which ensures current transfer through stud and not through the threaded portion of stud.
3	Insufficient bracing of CT secondary leads	May fail during primary short circuit condition	Mechanical	Secure supports are required for bracing of secondary leads
4	High dielectric stress in condenser core	Puncturing of primary graded insulation	Dielectric failure	To be checked during design review (Dielectric stress <4kV/mm)
5	Improper conductor insulation	Deteriorate under influence of high voltage stress	Dielectric failure	Check the incoming conductor insulation and also number of layers for conductor covering/grading

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6	Poor oil quality	High dielectric loss leading to PD and insulation failure.	Thermal and dielectric failure	Oil parameters to be as per relevant standards
7	Particles in oil held in suspension	PD and Temporary breakdown	Dielectric failure.	Maintain oil cleanliness and particle count as per relevant standards.
8	All metal components not earthed links not provided	Partial discharge may start and oil quality may get affected causing gassing.	Dielectric failure	All metal components are to be properly earthed and this is to be added in check-list
9	Tank bad & porous welding	Result in oil leakage and air/moisture ingress	Thermal/dielectric failure of primary insulation	Surface cleanliness to be ensured and adopt good welding procedures
10	Left over Metal parts inside the tank	Partial discharge	Dielectric damage to Primary Insulation due to Overheating of magnetic material in the high flux space	Better housekeeping/ supervision to be ensured
11	Sharp edges in Primary Aluminium conductors	Produce partial discharge and damage the conductor insulation	Dielectric failure	The surface finish should be smooth
12	Burrs on spacers and blocks	Result in damaging conductor insulation	Dielectric failure	Burr free condition to be ensured by good manufacturing facility

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13	Burrs on secondary core lamination	local short circuit and heating of core	Failure of CT secondary winding due to interturn fault.	Burr free condition to be ensured by good manufacturing facility
14	Improper drying process	Winding and insulation failure due to moisture	Thermal/dielectric failure	Drying and oil impregnation process to be maintained as per internal Quality standard.
15	Presence of sharp edges, voids, wrinkles in paper/foil layers	Incipient ionization leading to partial discharge	Dielectric failure	Supervision during manufacturing
16	Foreign material in insulation (dirt, debris etc)	Incipient ionization leading to partial discharge	Dielectric failure	Proper housekeeping and supervision during manufacturing.
17	PF test tap bad brazed joint	CT failure	Dielectric failure	Adopt good brazing procedures
18	Insulation surface contamination	insulation failure	Thermal/dielectric failure	Cleanliness to be ensured
19	Nitrogen leakage/ oil exposure	Moisture ingress, accelerated aging, insulation failure	Thermal/dielectric failure	Proper transportation/storage and erection at site
20	Leakage of oil during transportation /storage	Entry of air/moisture	Thermal/dielectric failure	Proper transportation/storage and erection at site

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21	Horizontal storage at site	Exposure of paper insulation	Damage to paperinsulation	Proper transportation/ storage and erection at site
22	Mechanical jerk during transportation	Shifting of support stubs for primary bar/secondary windings	Damage to paperinsulation	Proper transportation/storage and erection at site
23	Switching surges	Puncturing of primary graded insulation	Dielectric Failure	Design review
24	Lightning surges	Puncturing of primary graded insulation	Dielectric Failure	Design review
25	Transferred surges	Puncturing of primary graded insulation	Dielectric Failure	Design review
26	External short circuits with delayed fault clearance	Mechanical force in dead tank CT may lead to damage to primary insulation.	Dielectric Failure	Design review
27	Overloading of CT primary	Overheating of joints, primary bar	Thermal Failure	Amperage to be maintained within the rated capacity.
28	Oil leakage left unattended	Breathing of moisture and air into oil may cause increased dielectric heating	Thermal runaway	CTs with oil leakage to be promptly removed from service
29	Degraded Oil quality	High dielectric loss	Thermal runaway	CT to be removed from service
30	Slippage in Condition assessment	Unpredictable/unreliable CT performance	Could lead to disruptive failure	Monitoring as per OGN/OPS/ELEC/042
31	Slippage in ageing checks	Unpredictable/unreliable CT performance	Could lead to disruptive failure	Monitoring as per OGN/OPS/ELEC/042
32	Preventive maintenance	Unpredictable/unreliable CT performance	Could lead to disruptive failure	Monitoring as per OGN/OPS/ELEC/042

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	of critical accessories notdone			
33	CT PF test tap not earthed	Disruptive failure of CT	Dielectric failure	PF test tap earthing to be ensured with the help of a multimeter after each maintenance /testing
34	Deterioration of gasket material	Moisture entry through leaking joints, bellows	Thermal runaway	Replacement of gasket
35	Cracks in terminal board	CT oil leakage and air/moisture contamination	Dielectric failure	Removal of CT from service
36	Partial discharge in condenser core	Tracking over surface or burn-through of condenser core	Dielectric failure	Removal of CT from service
37	Oxygen entry through leaking joints, bellows	Accelerated ageing of oil, generation of PD	Thermal runaway and dielectric failure	Removal of CT from service
38	Over tightening of secondary stubs	Terminal board cracking causing oil leakage, air/moisture ingress	Dielectric failure	Care and supervision to be exercised during maintenance.
39	Looseness in primary terminal connections	Heating/hotspots at joint may cause damage to oil gasket and oil leakage / moisture ingress.	Dielectric heating causing thermal runaway.	To be attended promptly based on Thermal scanning results.

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40	Inadequate processing of CToil at site	Primary insulation failure	Thermal and dielectric	Processing of oil at site to be taken up under supervision of OEM and as per approved procedure.
41	Dirty porcelain petticoat	Cause external flashover during humid conditions / onset of rains especially in coastal regions. This may cause damage to insulator/tank/shell.	External Flashover	Porcelain petticoat to be cleaned at regular intervals. If phenomenon is more often hotline washing may be resorted to based on pollution monitoring.
42	Replacement of CT oil at site	Air entrapment causing primary Insulation failure	Dielectric	Work to be done with approved procedure
43	Replacement of secondary terminal plate at site	Air entrapment causing primary Insulation failure	Dielectric	Work to be done with approved procedure
44	Replacement of expansion bellows	Accidental dropping of foreign material inside the oil space	Dielectric damage to Primary insulation due to overheating of magnetic material in the high flux space	Due care and supervision to be exercised

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Annexure-3 DGA Evaluation Standards for Current Transformer

- ☐ IEC 60599 maximum admissible values for sealed instrument transformers without any action to be taken on the transformer (Values in microliters per liter)

H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂
300	300	900	30	50	10	2

- ☐ (CIGRE WG 12.16) Acceptance Criteria for oil Impregnated paper Insulated instrument Transformers

	Gas Concentration, ppm	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO	CO ₂
New Equipment	Acceptable Limit	<10	<5	<2	<5	none	<100	<200
In-Service Equipment	Do Nothing	<300	<30	<50	<10	<2	<300	<900
	Remove	>1000	>200	>500	>100	>10	No data	No data

TITLE	MONITORING THE HEALTHINESS OF 400KV CURRENT TRANSFORMER
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Annexure-4 MAINTENANCE SCHEDULE FOR CURRENT TRANSFORMER

01.	Checking of bellow expansion (level high/low/normal) and freeness of bellow expansion	Quarterly
02.	Visual inspection of CT for oil leakage/seepage from: • Dead tank gasket joint • Drain tap • Primary conductor sealing gasket Gauge glass • Metal bellows Schroeder valve Insulator cemented joint • Secondary terminal board/marshalling box	Monthly
03.	Thermo scanning of CT body and primary connections	Monthly
04.	Checking of primary connection strips, if provided externally	Yearly/OH
05.	Measurement of Tan and capacitance (C1)	Yearly/OH
06.	Tan delta tip up test	Yearly/OH
07.	Checking the tightness of PF terminal earthing connection. Check CT marshalling box sealing and gasket effectiveness.	Yearly/OH
08.	IR measurement (DAR)	Yearly/OH
09.	Checking of primary connection strips, if provided internally	SOS
10.	Measurement of CT secondary resistance	During OH
11.	Magnetization characteristics	SOS
12.	CT ratio test	SOS
13.	DGA and testing of other parameters of oil (Limits as specified in IEC60599)	Within warranty period and thereafter 10Y or SOS
14.	Checking of burden on the secondary winding	SOS
15.	On line acoustic PD measurement	SOS
16.	RVM test	SOS

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Annexure-5 ACCEPTANCE NORMS

CURRENT TRANSFORMERS		
(A)	IR Value by Insulation Tester	
	Primary – Earth	1000 M Ω (Min) by 5.0/10.0kV
	Secondary – Earth	50 M Ω (Min) by 0.5kV
	Control Cables	50 M Ω (Min) by 0.5kV
(B)	Tan delta value	0.007
		Since temperature correction factor for Tan-Delta depends on make, type and also aging conditions, the correction factors for different types/makes are different. Hence, no standard temp. error factors can be applied.
(C)	Rate of rise in Tan Delta	0.001 per year (max)
(D)	Rate of change in Capacitance	1% (max) per year
		Cumulative change not exceeding -5% and 10% from the Factory results.
(E)	Monitoring of Tan delta	Up to 0.007 (rise @ 0.001) 0.007 to 0.011 >0.011
		-Yearly/OH monitoring -Half yearly monitoring with Diagnostic of Oil and CT -replace the CT IEC 60599 (for limiting values)
(F)	Tan delta tip up test (Difference of tan delta measured at 2kV and 10kV)	0.001(max)
		Replace the CT
(G)	Terminal connector contact resistance	10 Micro-Ohm per connector
		NGC, UK Recommendations.
(H)	CT ratio errors	According to accuracy class on name plate
		IS-2705

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Annexure-6: DAILY MONITORING CHECKLIST OF CURRENT TRANSFORMERS :

BAY	OIL LEVEL			OIL LEAKAGE OBSERVED			ANY OTHER ABNORMALITY
	R PH	Y PH	B PH	R PH	Y PH	B PH	

Annexure-7: CURRENT TRANSFORMER -MONTHLY MAINTENANCE RECORD

Date of Commissioning		Make		Rating/Type		Sl.No.		Bay Location	
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MONTHLY MAINTENANCE - W/SD Activity

Visual inspection of CT for oil leakage and crack in insulators

Checking of bellow for expansion.....

MARSHALLING/ SECONDARY TERMINAL BOX

Check for any oil leakage from Secondary Terminal Box.....

Checking of healthiness of gaskets.....

Signature of Maintenance Engineer	
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Annexure- 8 : CURRENT TRANSFORMER – YEARLY MAINTENANCE RECORD

Date of Commissioning		Make		Rating/ Type		Sl. No.		Bay Location	
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- (a) Marshalling Box
- (b) Cleaning of MB.....
- (c) Checking the tightness of all electrical connections including earthing of MB
.....
- (d) Cleaning and tightness of CT secondary terminals and checking healthiness of sec terminal bushing
- (e) Checking of Space Heater. Sealing & Illumination.....
- (f) Thermo scanning of CT, Top dome & Body Kit used: Load:

	Ambient Temp.	Scanned Temp. R Phase	Scanned Temp. Y Phase	Scanned Temp. B Phase	Remarks
CT Tank					
Top Dome					
CT Body Temp					

Measurement of Capacitance & Tan δ

Make of the Kit.....Permissible Value of Tan δ is 0.007 at 20°C

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PHASE	Sl. No	Voltage Applied ,kV	Capacitance(C1/C2), pF		Tan δ , %		Remarks
			Pre-commg Values	Measured Value	Pre-commg Values	Measured Value	
R-Phase		2.0					Measurement to be taken after cleaning the porcelain surfaces • Results to be compared with Factory test/Pre commissioning values. Evaluate Tan delta tip up
		10.0					
Y-Phase		2.0					
		10.0					
B-Phase		2.0					
		10.0					

NOTE – TAN δ at 20 DEG C= 0.007(MAX), Rate of Rise of TAN δ /Year = 0.001(MAX), Rate of rise of Capacitance/Year = +/- 1 %(Max)

(i) IR Measurement (DAR)

Signature of Maintenance Engineer	
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Annexure-9: CT Equipment failure report

i.	Name of Substation	:	
ii.	Utility	:	
iii.	Faulty Equipment and the associated feeder	:	
iv.	Rating (Voltage/ current/ current ratio/ capacitance/ MCOV etc., as applicable)	:	

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v.	Short circuit withstand current and duration	:	
vi.	Make (Original equipment manufacturer)	:	
vii.	Serial No.	:	
viii.	Date and time of occurrence of fault	:	
ix.	Fault discovered during (Operation or periodic testing/ maintenance)	:	
x.	Weather conditions at the time of failure (clear sky/ rainy / thunderstorm etc.)		
xi.	Year of manufacturing	:	
xii.	Date of commissioning	:	
xiii.	Storage condition of equipment if the gap between delivery at site and commissioning is over 6 months	:	
xiv.	If OEM was contacted after failure, its recommendation/remark/ report/MoM	:	
xv.	Present condition of equipment (Repaired/ To be repaired / beyond repair)	:	
xvi.	(a) Details of previous maintenance (Activities carried out in previous maintenance including the tests conducted, periodicity of the maintenance activities) (b) Whether any abnormality observed in these tests. If yes, attach the test reports. (c) What steps were taken to address the abnormality?	:	
xvii.	Details of previous failure (Any failure which has led to a major repair of the equipment in the past since its commissioning)	:	
xviii.	Sequence of events/Description of fault (SOE with time stamp)	:	
xix.	Details of protection operated	:	
xx.	Details of tests done after failure	:	
xxi.	Observations (Any relevant observation made by the field staff or during visual inspection)	:	
xxii.	Probable cause of failure	:	
xxiii.	Attach reports of tests during pre-commissioning, periodic maintenance and after failure; Photographs of the failed equipment etc.	:	